Differential Manifolds

HHH

June 20, 2026

Contents

Chapter I

Manifolds

§1 Manifolds

§1.1 Topological Manifolds

Definition 1.1. *Suppose that M is a topological space. We say that M is a* ***topological manifold*** *of dimension n or a topological n-manifold if it has the following properties:*

(i) *M is a Hausdorff space*

(ii) *M is second-countable*

(iii) *M is locally Euclidean of dimension n: each point p of M has a neighborhood U that is homeomorphic to an open subset $\widehat{U}$ of $\mathbb{R}^n$. (U, φ) called* ***coordinate chart****.*

Given a chart (U, φ), we call the set U a ***coordinate domain****, and φ is called a* ***local coordinate map****.*

§1.2 Topological Properties of Manifolds

Definition 1.2. *Let M be a topological space.*

1. *A collection $\mathcal{X}$ of subsets of M is said to be* ***locally finite*** *if each point of M has a neighborhood that intersects at most finitely many of the sets in $\mathcal{X}$.*

2. *Given a cover $\mathcal{U}$ of M, another cover $\mathcal{V}$ is called a* ***refinement*** *of $\mathcal{U}$ if for each $V \in \mathcal{V}$ there exists some $U \in \mathcal{U}$ such that $V \subseteq U$.*

3. *We say that M is* ***paracompact*** *if every open cover of M admits an open, locally finite refinement.*

Proposition 1.3. *Suppose M is a topological manifold, then*

1. *local compact*

2. *σ-compact*

3. *M is Lindelof's : any open cover of M has a countable subcover.*

4. *paracompact*

Theorem 1.4 (Paracompact)**.** *Every topological manifold is paracompact. In fact, given a topological manifold M, an open cover $\mathcal{X}$ of M, and any basis $\mathfrak{B}$ for the topology of M, there exists a countable, locally finite open refinement of $\mathcal{X}$ consisting of elements of $\mathcal{B}$.*

Proof: Given $M, \mathcal{X}$, and $\mathcal{B}$ as in the hypothesis of the theorem, let $(K_j)_{j=1}^{\infty}$ be an exhaustion of M by compact sets (Proposition A.60). For each j, let $V_j = K_{j+1} \setminus \operatorname{Int} K_j$ and $W_j = \operatorname{Int} K_{j+2} \setminus K_{j-1}$ (where we interpret K_j as $\varnothing$ if $j < 1$). Then V_j is a compact set contained in the open subset W_j. For each $x \in V_j$, there is some $X_x \in \mathcal{X}$ containing x, and because $\mathcal{B}$ is a basis, there exists $B_x \in \mathcal{B}$ such that $x \in B_x \subseteq X_x \cap W_j$. The collection of all such sets B_x as x ranges over V_j is an open cover of V_j, and thus has a finite subcover. The union of all such finite subcovers as j ranges over the positive integers is a countable open cover of M that refines $\mathcal{X}$. Because the finite subcover of V_j consists of sets contained in W_j, and $W_j \cap W_{j'} = \varnothing$ except when $j-2 \leq j' \leq j+2$, the resulting cover is locally finite.

§1.3 Smooth Structures and Smooth Manifolds

Definition 1.5. *Let M be a topological n-manifold.*

1. *Two charts (U, φ) and (V, ψ) are said to be* ***smoothly compatible*** *if either $U \cap V = \varnothing$ or the two* ***transition map***

$$\psi \circ \varphi^{-1} : \varphi(U \cap V) \to \psi(U \cap V), \quad \varphi \circ \psi^{-1} : \psi(U \cap V) \to \varphi(U \cap V)$$

   are C^∞.

2. *We define an* ***smooth atlas*** *$\mathcal{A}$ for M to be a collection of charts $\{(U_\alpha, \varphi_\alpha)\}$ whose domains cover M and any two charts in $\mathcal{A}$ are smoothly compatible with each other.*

3. *A smooth atlas $\mathcal{A}$ on M is* ***maximal*** *if it is not properly contained in any larger smooth atlas. If M is a topological manifold, a* ***smooth structure*** *on M is a maximal smooth atlas.*

   A C^∞ manifold is a pair $(M, \mathcal{A})$, where M is a topological manifold and $\mathcal{A}$ is a ***smooth structure*** *on M.*

Although the C^∞ compatibility of charts is clearly reflexive and symmetric, it is not transitive. However, we have the following lemma:

Lemma 1.6. *Let $\{(U_\alpha, \phi_\alpha)\}$ be an atlas on a locally Euclidean space. If two charts (V, ψ) and (W, σ) are both compatible with the atlas $\{(U_\alpha, \phi_\alpha)\}$, then they are compatible with each other.*

Proposition 1.7. *Let M be a topological manifold.*

1. *Every smooth atlas $\mathcal{A}$ for M is contained in a unique maximal smooth atlas $\overline{\mathcal{A}}$, called the* ***smooth structure determined by*** *$\mathcal{A}$. Indeed, $\overline{\mathcal{A}}$ denote the set of all charts that are smoothly compatible with every chart in $\mathcal{A}$.*

2. *Two smooth atlases for M determine the same smooth structure if and only if their union is a smooth atlas.*

§1.4 Manifold with Boundary

Denote by $\mathbb{R}_+^n$ or $\mathbb{H}^n$ the subset of $\mathbb{R}^n$ defined by

$$\mathbb{R}_+^n = \left\{ \left(x^1, \ldots, x^n\right) \in \mathbb{R}^n : x^n \geq 0 \right\}$$

Definition 1.8. *An* ***n-dimensional topological manifold with boundary*** *is a second-countable Hausdorff space M in which every point p has a neighborhood U homeomorphic φ to an open subset of $\mathbb{R}_+^n$.*

1. *The open subset $U \subseteq M$ together with a map $\varphi : U \to \mathbb{R}_+^n$ that is a homeomorphism onto an open subset of $\mathbb{R}_+^n$ will be called a* ***chart*** *for M.*

2. *We will call (U, φ) an* ***interior chart*** *if $\varphi(U)$ is an open subset of $\mathbb{R}^n$, and a* ***boundary chart*** *if $\varphi(U) \cap \partial\mathbb{R}_+^n \neq \varnothing$.*

3. *A point $p \in M$ is called* ***interior point*** *if it is in the domain of some interior chart.*

   A point $p \in M$ is called ***boundary point*** *if it is in the domain of a boundary chart that sends p to $\partial\mathbb{R}_+^n$.*

The boundary of M (the set of all its boundary points) is denoted by ∂M; similarly, its interior, the set of all its interior points, is denoted by $\operatorname{Int} M$.

Theorem 1.9. *Each point of M is either an interior point or a boundary point, but not both.*

Corollary 1.10. *Let M be a topological n-manifold with boundary.*

1. $\operatorname{Int} M$ *is an open subset of M and a topological n-manifold without boundary.*

2. *∂M is a closed subset of M and a topological $(n-1)$-manifold without boundary.*

3. *M is a topological manifold if and only if $\partial M = \varnothing$.*

4. *If $n = 0$, then $\partial M = \varnothing$ and M is a 0 -manifold.*

Smooth Manifold with Boundary

Recall that a map from an arbitrary subset $A \subseteq \mathbb{R}^n$ to $\mathbb{R}^k$ is said to be smooth if in a neighborhood of each point of A it admits an extension to a smooth map defined on an open subset of $\mathbb{R}^n$.

Definition 1.11. *Let M be a topological manifold with boundary. A **smooth structure** for M is defined to be a maximal smooth atlas $\mathcal{A}$, a collection of charts whose domains cover M and whose transition maps (and their inverses) are smooth. With such a structure, $(M, \mathcal{A})$ is called a **smooth manifold with boundary**.*

§2 Smooth Maps on a Manifold

Definition 2.1. *Let M be a manifold, k is a nonnegative integer, and $f : M \to \mathbb{R}^k$ is any function. We say that f is **smooth at** p if there exists a chart (U, φ) for M whose domain contains p and such that the composite function*

$$f \circ \varphi^{-1} : \varphi(U) \to \mathbb{R}^k$$

*is smooth at $\varphi(p)$. The function $f \circ \varphi^{-1}$ is called the **local coordinate representation** of f. The function f is said to be C^∞ on M if it is C^∞ at every point of M.*

Remark. *The definition of the smoothness of a function f at a point is independent of the chart (U, φ).*

Proposition 2.2. *Let M be a manifold of dimension n, and $f : M \to \mathbb{R}$ a real-valued function on M. The following are equivalent:*

1. *The function $f : M \to \mathbb{R}$ is C^∞.*
2. *The manifold M has an atlas such that for every chart (U, ϕ) in the atlas, $f \circ \phi^{-1} : \mathbb{R}^n \supset \phi(U) \to \mathbb{R}$ is C^∞.*
3. *For every chart (V, ψ) on M, the function $f \circ \psi^{-1} : \mathbb{R}^n \supset \psi(V) \to \mathbb{R}$ is C^∞.*

Proposition 2.3 (Smoothness in terms of components)**.** *Let M be a manifold. A vector-valued function $F : M \to \mathbb{R}^m$ is C^∞ if and only if its component functions $F^1, \ldots, F^m : M \to \mathbb{R}$ are all C^∞.*

Definition 2.4. *Let M, N be smooth manifolds, and let $f : M \to N$ be a continuous map. We say that f is a **smooth map** if for every $p \in M$, there exist smooth charts (U, φ) containing p and (V, ψ) containing $f(p)$ (assume that $f(U) \subseteq V$ without generality) and the composite map*

$$\tilde{f} = \psi \circ f \circ \varphi^{-1} : \varphi(f^{-1}(V) \cap U) \to \psi(V)$$

*is smooth. We call $\tilde{f} = \psi \circ f \circ \varphi^{-1}$ the **coordinate representation of** f **with respect to the given coordinates**.*

Proposition 2.5. *Let N and M be smooth manifolds, and $F : N \to M$ a continuous map. The following are equivalent:*

1. *The map $F : N \to M$ is C^∞.*
2. *There are atlases $\mathfrak{U}$ for N and $\mathfrak{V}$ for M such that for every chart (U, ϕ) in $\mathfrak{U}$ and (V, ψ) in $\mathfrak{V}$, the map*
$$\psi \circ F \circ \phi^{-1} : \phi\left(U \cap F^{-1}(V)\right) \to \mathbb{R}^m$$
is C^∞.
3. *For every chart (U, ϕ) on N and (V, ψ) on M, the map*
$$\psi \circ F \circ \phi^{-1} : \phi\left(U \cap F^{-1}(V)\right) \to \mathbb{R}^m$$
is C^∞.

Proposition 2.6 (Composition of C^∞ maps)**.** *Let M, N, and P be smooth manifolds with or without boundary.*

1. *Every constant map $M \to N$ is smooth.*
2. *The identity map of M is smooth.*
3. *If $U \subseteq M$ is an open submanifold with or without boundary, then the inclusion map $U \hookrightarrow M$ is smooth.*
4. *If $F : M \to N$ and $G : N \to P$ are smooth, then so is $G \circ F : M \to P$.*

Definition 2.7. *If M and N are smooth manifolds with or without boundary, a **diffeomorphism** from M to N is a smooth bijective map $F : M \to N$ that has a smooth inverse. We say that M and N are **diffeomorphic** if there exists a diffeomorphism between them. Sometimes this is symbolized by $M \approx N$.*

Proposition 2.8. *Let U be an open subset of a manifold M of dimension n and $f : U \to f(U) \subset \mathbb{R}^n$ be a maps. Then f is a diffeomorphism onto its image if and only if (U, f) is a coordinate chart in the differentiable structure of M.*

§3 Partition of unity

§3.1 Topological Preliminaries

Definition 3.1. *Let X be a topological space, if there exists X_j such that*

(i) *For each j, the closure $\overline{X}_j$ is compact.*

(ii) *For each j, $\overline{X}_j \subset X_{j+1}$.*

(iii) $M = \bigcup_j X_j$.

The subsets $\{X_j\}$ *described is called an* ***exhaustion*** *of* M.

Definition 3.2. *A real-valued continuous function* f *on* X *is called an* ***exhausion function*** *for* X *if for any* $c \in \mathbb{R}$*, the sublevel set* $f^{-1}((\infty, c])$ *is compact.*

Lemma 3.3 (Lindelof's)**.** *Let* X *be a second countable space, then every open cover* $\mathcal{A}$ *has a countable subcover.*

Theorem 3.4. *If* X *is a second countable, locally compact and Hausdorff space (thus a manifold), then there exists a exhaustion of* X.

Definition 3.5. *Let* $\mathcal{A}$ *be a open cover of the space* X*, then a open cover* $\mathcal{B}$ *of* X *is said to be a* ***refinement*** *of* $\mathcal{A}$ *(or is said to refine* $\mathcal{A}$ *) if for each element* B *of* $\mathcal{B}$*, there is an element* A *of* $\mathcal{A}$ *containing* B.

Definition 3.6. *A space* X *is* ***paracompact*** *if every open cover* $\mathcal{A}$ *of* X *has a locally finite open refinement* $\mathcal{B}$ *of* $\mathcal{A}$.

Theorem 3.7. *Let* M *be any topological manifold. For any open cover* $\mathcal{U} = \{U_\alpha\}$ *of* M*, one can find two countable family of open covers* $\mathcal{V} = \{V_j\}$ *and* $\mathcal{W} = \{W_j\}$ *of* M *so that*

(i) $\mathcal{W}$ *is a locally finite open refinement of* $\mathcal{U}$.

(ii) For each j, $\overline{V}_j$ *is compact and* $\overline{V}_j \subset W_j$.

Proof. *Let* $\{X_j\}$ *be a exhaustion of* M*. For each* $p \in M$*, there is an* j *and an* $\alpha(p)$ *so that* $p \in \overline{X}_{j+1} \backslash X_j$ *and* $p \in U_{\alpha(p)}$*. Since* M *is locally Euclidean, one can always choose open neighborhoods* V_p, W_p *of* p *so that* $\overline{V}_p$ *is compact and*

$$p \in V_p \subset \overline{V}_p \subset W_p \subset U_{\alpha(p)} \cap \left(X_{j+2} \backslash \overline{X}_{j-1}\right)$$

Now for each j*, since the "stripe"* $\overline{X}_{j+1} \backslash X_j$ *is compact, one can choose finitely many points* $p_1^j, \cdots, p_{k_j}^j$ *so that* $V_{p_1^j}, \cdots, V_{p_{k_j}^j}$ *is an open cover of* $\overline{X}_{j+1} \backslash X_j$*. Denote all these* $V_{p_k^j}$ *'s by* $V_1, V_2, \cdots$*, and the corresponding* $W_{p_k^j}$ *'s by* $W_1, W_2, \cdots$*. Then* $\mathcal{V} = \{V_k\}$ *and* $\mathcal{W} = \{W_k\}$ *are open covers of* M *that satisfies all the conditions.*

Corollary 3.8. *Manifolds is paracompact,* σ*-compact*

§3.2 Existence

Lemma 3.9. *There exists* f_1, f_2, f_3 *such that*

Theorem 3.10 (Bump function on manifold)**.** *Let M be a smooth manifold, $K \subset M$ is a compact subset, and $U \subset M$ an open subset that contains K. Then there is a $\varphi \in C^\infty(M)$ so that $K \prec f \prec U$ ($0 \leq \varphi \leq 1, \varphi \equiv 1$ on K and $\operatorname{supp}(\varphi) \subset U$).*

Proof. For each $q \in K$, there is a chart (φ_q, U_q, V_q) near q so that $U_q \subset U$ and V_q contains the open ball $B_3(0)$ in $\mathbb{R}^n$. Let $U'_q = \varphi_q^{-1}(B_1(0))$, and let

$$f_q(p) = \begin{cases} f_3(\varphi_q(p)) & , p \in U_q \\ 0 & , p \notin U_q \end{cases}$$

Then $f_q \in C^\infty(M), \operatorname{supp}(f_q) \subset U_q \subset U$ and $f_1 \equiv 1$ on U'_q.

Now the family of open sets $\{U'_q\}$ is an open cover of K. Since K is compact, there is a finite sub-cover $\{U'_{q_1}, \cdots, U'_{q_n}\}$. Let

$$\psi = \sum_{i=1}^{n} f_{q_i}$$

Then ψ is a smooth and compactly supported function on M so that $\psi \geq 1$ on K and $\operatorname{supp}(\psi) \subset U$. It follows that the function $\varphi(p) = f_2(\psi(p))$ satisfies all the conditions we required.

Definition 3.11. *Suppose M is a topological space, and let $\mathcal{A} = \{A_\alpha\}$ be an arbitrary open cover of M. A* ***partition of unity subordinate to*** *$\mathcal{A}$ is an indexed family*

$$\{\rho_\alpha : \rho_\alpha : M \to \mathbb{R} \text{ is continue}\}$$

with the following properties:

(i) $\rho_\alpha \prec A_\alpha$ ($\operatorname{supp} \rho_\alpha \subset A_\alpha$) *for all α*

(ii) The family of supports $\{\operatorname{supp} \rho_\alpha\}$ is locally finite.

(iii) $\sum_\alpha \rho_\alpha(x) \equiv 1$ *on M*

Theorem 3.12. *Suppose M is a smooth manifold with or without boundary, and $\{U_\alpha\}_{\alpha \in A}$ is any indexed open cover of M. Then there exists a smooth partition of unity subordinate to $\{U_\alpha\}_{\alpha \in A}$.*

Proof. We can find nonnegative functions $\varphi_j \in C^\infty(M)$ so that $\overline{V}_j \prec \varphi_j \prec W_j$ on $\overline{V}_j$ and $\operatorname{supp}(\varphi_j) \subset W_j$. *Since $\mathcal{W}$ is a locally finite cover, $\varphi = \sum \varphi_j$ is a well-defined smooth function on M. Since each φ_j is nonnegative, and $\mathcal{V}$ is a cover of M, φ is strictly positive on M. It follows that the functions $\psi_j = \frac{\varphi_j}{\varphi}$ are smooth and satisfy $0 \leq \psi_j \leq 1$ and $\sum_j \psi_j = 1$.*

Let

$$\rho_\alpha = \sum_{W_j \subset U_\alpha} \psi_j$$

Note that the right hand side is a finite sum near each point, so it does define a smooth function. Clearly the family $\{\rho_\alpha\}$ is a partition of unity subordinate to $\{U_\alpha\}$.

§3.3 Application

Theorem 3.13 (Smooth Urysohn lemma)**.** *Let M be a manifold, $F \subset M$ is a closed subset, and $U \subset M$ an open subset that contains F. Then there is a "bump" function $\varphi \in C^\infty(M)$ so that $0 \leq \varphi \leq 1, \varphi \equiv 1$ on F and* $\operatorname{supp}(\varphi) \subset U$.

Proof. *Let $\{\rho_1, \rho_2\}$ be a partition of unity subordinate to the open cover $\{U, M \backslash F\}$. Then $\varphi = \rho_1$ is what we need: $\rho_1 = 1$ on F since $\rho_2 = 0$ on F.*

Theorem 3.14 (Whitney Approximation Theorem)**.** *Let M be a smooth manifold, $F \subset M$ a closed subset and k be a positive integer. Then for any continuous function $f : M \to \mathbb{R}^k$ which is smooth on F and any positive continuous function $\delta : M \to \mathbb{R}_{>0}$, there exists $f \in C^\infty(M)$ so that*

$$f(p) = g(p), \quad \forall p \in F$$

and

$$|f(p) - g(p)| < \delta(p), \quad \forall p \in M$$

Proof. *By definition, there exists an open set $U \supset A$ and a smooth function f_0 defined on U so that $f_0 = f$ on F. Let*

$$U_0 = \{p \in U : |f_0(p) - f(p)| < \delta(p)\}$$

Then U_0 is open in M and $U_0 \supset F$.

Next we construct a open cover of $M \backslash F$. For any $q \in M \backslash F$, we let

$$U_q = \{p \in M \backslash A : |f(p) - f(q)| < \delta(p)\}$$

Then $\{U_q \mid q \in M \backslash F\}$ is an open covering of $M \backslash F$. Now let $\{\rho_0, \rho_q : q \in M\}$ be P.O.U. subordinate to the open cover $\{U_0, U_q : q \in M\}$ of M, and define a function on M via

$$g(p) = \rho_0(p) f_0(p) + \sum_{q \in M} \rho_q(p) f(q).$$

Since the summation is locally finite, g is smooth. Also by definition, $g = f_0 = f$ on F. Moreover, for any $q \in M$ one has

$$\begin{aligned}
|g(p) - f(p)| &= \left| \rho_0(p) f_0(p) + \sum_q \rho_q(p) f(q) - \rho_0(p) f(p) - \sum_q \rho_q(p) f(p) \right| \\
&\leq \rho_0(p) \left| f_0(p) - f(p) \right| + \sum_q \rho_q(p) |f(q) - f(p)| \\
&< \rho_0(p) \delta(p) + \sum_q \rho_q(p) \delta(p) \\
&= \delta(p)
\end{aligned}$$

Corollary 3.15 (Tietze)**.** *Let M be a smooth manifold and closed set $F \subset$. If f is smooth on F,* *then there exists $g \in C^{\infty}(M)$ that $f = g$ on F.*

Theorem 3.16 (Existence of Smooth Exhaustion Function)**.** *Every smooth manifold with or without boundary admits a smooth positive exhaustion function.*

Proof. Let M be a smooth manifold with or without boundary, let $\{V_j\}_{j=1}^{\infty}$ be any countable open cover of M by precompact open subsets, and let $\{\psi_j\}$ be a smooth partition of unity subordinate to this cover. Define $f \in C^{\infty}(M)$ by

$$f(p) = \sum_{j=1}^{\infty} j\psi_j(p)$$

Then f is smooth because only finitely many terms are nonzero in a neighborhood of any point, and positive because $f(p) \geq \sum_j \psi_j(p) = 1$.

To see that f is an exhaustion function, let $c \in \mathbb{R}$ be arbitrary, and choose a positive integer $N > c$. If $p \notin \bigcup_{j=1}^{N} \bar{V}_j$, then $\psi_j(p) = 0$ for $1 \leq j \leq N$, so

$$f(p) = \sum_{j=N+1}^{\infty} j\psi_j(p) \geq \sum_{j=N+1}^{\infty} N\psi_j(p) = N\sum_{j=1}^{\infty} \psi_j(p) = N > c.$$

Equivalently, if $f(p) \leq c$, then $p \in \bigcup_{j=1}^{N} \bar{V}_j$. Thus $f^{-1}((-\infty, c])$ is a closed subset of the compact set $\bigcup_{j=1}^{N} \bar{V}_j$ and is therefore compact. □

Theorem 3.17 (Level Sets of Smooth Functions)**.** *Let M be a smooth manifold. If K is any closed subset of M, there is a smooth nonnegative function $f : M \to \mathbb{R}$ such that $f^{-1}(0) = K$.*

Proof. We begin with the special case in which $M = \mathbb{R}^n$ and $K \subseteq \mathbb{R}^n$ is a closed subset. For each $x \in M \backslash K$, there is a positive number $r \leq 1$ such that $B_r(x) \subseteq M \backslash K$. By Proposition A.16, $M \backslash K$ is the union of countably many such balls $\{B_{r_i}(x_i)\}_{i=1}^{\infty}$.

Let $h : \mathbb{R}^n \to \mathbb{R}$ be a smooth bump function that is equal to 1 on $\bar{B}_{1/2}(0)$ and supported in $B_1(0)$. For each positive integer i, let $C_i \geq 1$ be a constant that bounds the absolute values of h and all of its partial derivatives up through order i. Define $f : \mathbb{R}^n \to \mathbb{R}$ by

$$f(x) = \sum_{i=1}^{\infty} \frac{(r_i)^i}{2^i C_i} h\left(\frac{x - x_i}{r_i}\right)$$

The terms of the series are bounded in absolute value by those of the convergent series $\sum_i 1/2^i$, so the entire series converges uniformly to a continuous function by the Weierstrass M-test. Because the i th term is positive exactly when $x \in B_{r_i}(x_i)$, it follows that f is zero in K and positive elsewhere.

It remains only to show that f is smooth. We have already shown that it is continuous, so suppose $k \geq 1$ and assume by induction that all partial derivatives of f of order less than k exist and are continuous. By the chain rule and induction, every k th partial derivative of the i th term in the series can be written in the form

$$\frac{(r_i)^{i-k}}{2^i C_i} D_k h\left(\frac{x-x_i}{r_i}\right),$$

where $D_k h$ is some k th partial derivative of h. By our choices of r_i and C_i, as soon as $i \geq k$, each of these terms is bounded in absolute value by $1/2^i$, so the differentiated series also converges uniformly to a continuous function. It then follows from Theorem C. 31 that the k th partial derivatives of f exist and are continuous. This completes the induction, and shows that f is smooth.

Now let M be an arbitrary smooth manifold, and $K \subseteq M$ be any closed subset. Let $\{B_\alpha\}$ be an open cover of M by smooth coordinate balls, and let $\{\psi_\alpha\}$ be a subordinate partition of unity. Since each B_α is diffeomorphic to $\mathbb{R}^n$, the preceding argument shows that for each α there is a smooth nonnegative function $f_\alpha : B_\alpha \to \mathbb{R}$ such that $f_\alpha^{-1}(0) = B_\alpha \cap K$. The function $f = \sum_\alpha \psi_\alpha f_\alpha$ does the trick. □

§3.4 In paracompact Hausdorff space

Lemma 3.18 (Shrinking lemma)**.** *Let X be a paracompact Hausdorff space; let $\{U_\alpha\}_{\alpha\in J}$ be open cover of X. Then there exists a locally finite open cover $\{V_\alpha\}_{\alpha\in J}$ of X such that $\overline{V}_\alpha \subset U_\alpha$ for each α.*

Proof. Let $\mathcal{A}$ be the collection of all open sets A such that $\overline{A}$ is contained in some element of the collection $\{U_\alpha\}$. Regularity of X implies that $\mathcal{A}$ covers X. Since X is paracompact, we can find a locally finite collection $\mathcal{B} = \{B_\beta\}_{\beta\in K}$ of open sets covering X that refines $\mathcal{A}$

Let us index $\mathcal{B}$ bijectively with some index set K, then the general element of $\mathscr{B}$ can be denoted B_β, for $\beta \in K$, and $\{B_\beta\}_{\beta\in K}$ is a locally finite indexed famuly. Since $\mathscr{B}$ refines $\mathscr{A}$, we can define a function $f : K \to J$ by choosing, for each β in K, an element $f(\beta) \in J$ such that

$$\overline{B}_\beta \subset U_{f(\beta)}$$

Then for each $\alpha \in J$, we define V_α to be the union of many B_β

$$V_\alpha = \bigcup_{f(\beta)=\alpha} B_\beta$$

Because the collection $\mathcal{B}_\alpha$ is locally finite, $\overline{V}_\alpha = \bigcup \overline{B}_\beta$, so that $\overline{V}_\alpha \subset U_\alpha$.

Finally, we check local finiteness. Given $x \in X$, choose a neighborhood W of x such that W intersects B_β for only finitely many values of β, say $\beta = \beta_1, \ldots, \beta_K$. Then W can intersect V_α only if α is one of the indices $f(\beta_1), \ldots, f(\beta_K)$.

Theorem 3.19. *Let X be a paracompact Hausdorff space; let $\{U_\alpha\}_{\alpha\in J}$ be an indexed open covering of X. Then there exists a partition of unity on X subordinate to $\{U_\alpha\}$.*

Proof. We begin by applying the shrinking lemma twice, to find locally finite indexed famles of *open sets $\{W_\alpha\}$ and $\{V_\alpha\}$ covering X, such that*

$$W_\alpha \subset \overline{W}_\alpha \subset V_\alpha \subset \overline{V}_\alpha \subset U_\alpha$$

for each α Since X is normal, we may choose, for each α, a continuous function $\varphi_\alpha : X \to [0,1]$ such that $\varphi_\alpha\left(\overline{W}_\alpha\right) = \{1\}$ and $\varphi_\alpha(X - V_\alpha) = \{0\}$. Since φ_α is nonzero only at points of V_α, we have

$$\operatorname{supp} \varphi_\alpha \subset \overline{V}_\alpha \subset U_\alpha$$

Furthermore, the indexed family $\left\{\overline{V}_\alpha\right\}$ is locally finite (since an open set intersects $\overline{V}_\alpha$ only if it intersects V_α); hence the indexed family supp φ_α *is also locally finite. Note that because $\{W_\alpha\}$ covers X, for any given x at least one of the functions ψ_α is positive at x.*

We can now make sense of the formally infinite sum

$$\varphi(x) = \sum_\alpha \varphi_\alpha(x)$$

and define

$$\rho_\alpha(x) = \frac{\varphi_\alpha(x)}{\varphi(x)}$$

to obtain our desired partition of unity.

§4 Sard's Theorem

Definition 4.1. *We say that a subset $A \subseteq M$ **has measure zero in** M if for every smooth chart (U, φ) for M, the subset $\varphi(A \cap U) \subseteq \mathbb{R}^n$ has n-dimensional measure zero.*

Lemma 4.2. *Let M be a smooth n-manifold with or without boundary and $A \subseteq M$. Suppose that for some collection $\{(U_\alpha, \varphi_\alpha)\}$ of smooth charts whose domains cover $A, \varphi_\alpha(A \cap U_\alpha)$ has measure zero in $\mathbb{R}^n$ for each α. Then A has measure zero in M.*

Theorem 4.3. *Suppose M and N are differential m-manifolds with or without boundary, $F : M \to N$ is a C^1 map, and $A \subseteq M$ is a subset of measure zero. Then $F(A)$ has measure zero in N.*

Corollary 4.4. *Suppose M and N are differential manifolds with or without boundary,* $\dim M \leq \dim N$*, and $F \in C^1(M, N)$. If $A \subset M$ is a subset of measure zero, then $F(A)$ has measure zero in N.*

Definition 4.5. *If $f : M \to N$ is a smooth map,*

*(1) A point $p \in M$ is said to be a **regular point of** f if $df_p : T_pM \to T_{f(p)}N$ is surjective* ($\operatorname{rank}_p f = \dim N$). *A point $c \in N$ is said to be a **regular value** of f if every point of the level set $f^{-1}(c)$ is a regular point.*

*(2) A point $q \in M$ is said to be a **critical point of** f if $df_p : T_pM \to T_{f(p)}N$ is not surjective* ($\operatorname{rank}_p f < \dim N$). *A point $d \in N$ is said to be a **critical value** of f if there exists a critical point in level set $f^{-1}(d)$.*

Theorem 4.6 (Sard's Theorem)**.** *Suppose M and N are manifolds, and $f : M \to N$ is a smooth map. Then the set of critical values of f has measure zero in N.*

Corollary 4.7. *Suppose M and N are smooth manifolds with or without boundary, and $F : M \to N$ is a smooth map. If $\dim M < \dim N$, then $F(M)$ has measure zero in N.*

Corollary 4.8. *Suppose M is a smooth manifold with or without boundary, and $S \subseteq M$ is an immersed submanifold with or without boundary. If $\dim S < \dim M$, then S has measure zero in M.*

Chapter II

Tangent and Cotangent Spaces

Contents

§1 Basic Definitions

Definition 1.1. *Let M be a manifold, p be a point in M.*

1. *The* ***germ*** *of a C^∞ function at p in M is defined*

$$C_p^\infty(M) := \varinjlim_{U \ni p} C^\infty(U)$$

2. *A* ***tangent vector*** *at p is a derivation of $\mathbb{R}$-algebra $C_p^\infty(M)$ (a linear map $D : C_p^\infty(M) \to \mathbb{R}$ such that $D(fg) = (Df)g(p) + f(p)Dg$)*

3. *The tangent vectors at p form a vector space*

$$T_pM := \mathrm{Der}_{\mathbb{R}}\left(C_p^\infty(M), \mathbb{R}\right)$$

called the ***tangent space*** *of M at p.*

4. *A* ***cotangent vector*** *at p is a linear functional on the tangent space T_pM. And the* ***cotangent space*** *of M at p is the dual vector space*

$$T_p^*M := \mathrm{Hom}_{\mathbb{R}}\left(T_pM, \mathbb{R}\right)$$

Definition 1.2. *We define the* ***tangent bundle*** *of M:*

$$TM := \coprod_{p \in M} T_pM$$

Remark. *We usually write an element of this disjoint union as an ordered pair (p, X) of X_p, with $p \in M$ and $X \in T_pM$. The tangent bundle comes equipped with a* ***natural projection map***

$$\pi : TM \to M$$

which sends each vector in T_pM to the point p at which it is tangent: $\pi(p, X) = p$.

Proposition 1.3. *The tangent bundle TM has a natural topology and smooth structure that make it into a $2n$-dimensional smooth manifold. With respect to this structure, the projection π is smooth.*

Definition 1.4. *The cotan*

§2 The Differential of a Smooth Map

Definition 2.1. *Let $f : M \to N$ be smooth map between manifolds and a point $p \in M$. We define a map induced by f on the tangent spaces*

$$\mathrm{d}f_p : T_pM \to T_{f(p)}N$$

as follows. Given $v \in T_pM$, we let $\mathrm{d}f_p(v)$ be the derivation at $f(p)$ that acts on $C^\infty_{f(p)}(N)$ by the rule

$$\langle \mathrm{d}f_p(v), \varphi\rangle = \langle v, \varphi \circ f\rangle \quad \text{for all } \varphi \in C^\infty_{f(p)}(N)$$

Proposition 2.2. *There is a smooth map called the* ***pushforward*** *of f*

$$f_* : TM \to TN, \quad (p, v) \mapsto (f(p), \mathrm{d}f_p(v))$$

Proposition 2.3. *Tangent functor T :* **Man** $\to$ **VectBun** *is covariant*

§2.1 Computations in Coordinates

Given a chart (U, φ, x^i) covering a point $p \in M$, $\mathrm{d}\varphi_p$ is an isomorphism between T_pM and $T_p\mathbb{R}^n$. Thus we can choose a basis of T_pM by

$$\left.\frac{\partial}{\partial x^i}\right|_p := (\mathrm{d}\varphi_p)^{-1}\left(\left.\frac{\partial}{\partial x^i}\right|_{\hat{p}}\right)$$

Proposition 2.4. *Given a smooth map $f : M \to N$, and charts (U, φ, x^i) and (V, ψ, y^j) for M and*

N respectively, then

$$df_p\left(\left.\frac{\partial}{\partial x^i}\right|_p\right)=\sum_j\frac{\partial y^j}{\partial x^i}(\hat{p})\cdot\left.\frac{\partial}{\partial y^j}\right|_{f(p)}$$

where $(y^1,\ldots,y^n)=\psi\circ f\circ\varphi^{-1}(x^1,\ldots,x^n)$. Thus the representation matrix of df_p with respect to the bases $\left\{\left.\frac{\partial}{\partial x^i}\right|_p\right\}$ and $\left\{\left.\frac{\partial}{\partial y^j}\right|_{f(p)}\right\}$ is the Jacobian matrix of $\tilde{f}$ at $\hat{p}$.

Corollary 2.5. *Suppose that (U,x^i) and $(\tilde{U},\tilde{x}^i)$ are smooth charts for M centered at p, then*

$$df_p\left(\left.\frac{\partial}{\partial x^i}\right|_p\right)=\frac{\partial\tilde{x}^j}{\partial x^i}(\hat{p})\cdot\left.\frac{\partial}{\partial\tilde{x}^j}\right|_p$$

§3 Vector fields

Definition 3.1. *A **vector field** on M is a section of $\pi:TM\to M$, i.e. a map $X:M\to TM$ such that $\pi\circ X=\mathrm{id}_M$. The **support** of a vector field X is the closure of the set $\{p\in M:X(p)\neq 0\}$. We say that X is a **smooth vector field** if it is smooth as a map from M to TM.*

Remark. *If (U,x^i) is a smooth chart, we can write X (omit the base point) in coordinates locally as*

$$X(p)=\sum_i X^i(p)\cdot\left.\frac{\partial}{\partial x^i}\right|_p$$

*This n functions $X^i:U\to\mathbb{R}$ are called the **components funtions** of X in the given chart.*

Proposition 3.2 (Smoothness Criterion for Vector Fields)**.** *A vector field X is smooth if and only if its component functions are smooth.*

Proposition 3.3. *The set of all smooth vector fields on M is a $C^\infty(M)$-module:*

$$\mathfrak{X}(M)=\mathrm{Der}_{\mathbb{R}}\left(C^\infty(M),C^\infty(M)\right)$$

Definition 3.4. *Let $f:M\to N$ be a smooth map, and let $X\in\mathfrak{X}(M)$. The vector field X and Y are said to be **f-related** if for every $p\in M$,*

$$f_{*,p}(X_p)=Y_{f(p)}$$

Proposition 3.5. *If f is a diffeomorphism, then*

1. *for every $X\in\mathfrak{X}(M)$, there is a unique vector field Y on N that is f-related to X called the **pushforward** of X under f. Explicitly,*

$$(f_*X)_q=f_{*,f^{-1}(q)}(X_{f^{-1}(q)})$$

§3.1 Integral Curves

Given a smooth curve $\gamma : J \to M$ and $t_0 \in J$, we define the **velocity** of γ at t_0, denoted by $\gamma'(t_0)$, to be the vector

$$\gamma'(t_0) := \gamma_{*,t_0}\left(\left.\frac{d}{dt}\right|_{t_0}\right) \in T_{\gamma(t_0)}M$$

where $d/\left.dt\right|_{t_0}$ is the standard coordinate basis vector in $T_{t_0}\mathbb{R}$.

Definition 3.6. *Let $X \in \mathfrak{X}(M)$ be a smooth vector field. The curve $\gamma : J \to M$ is called an **integral curve** of X if $\gamma'(t) = X_{\gamma(t)}$ for every $t \in J$*

§3.2 Flow

Definition 3.7. *A **global flow** on M is a left $\mathbb{R}$-action $\theta : \mathbb{R} \times M \to M$.*

Definition 3.8. *A **flow** on M is a continuous map $\theta : \mathcal{D} \to M$*

- *(i) $\mathcal{D} \subseteq \mathbb{R} \times M$ is the **flow domain** for M, that is, $\mathcal{D}^{(p)} := \{t \in \mathbb{R} | (p,t) \in \mathcal{D}\}$ is an open interval containing 0 for each $p \in M$*
- *(ii) $\theta(0,p) = p$ for all $p \in M$, and*
- *(iii) for all $s \in \mathcal{D}^{(p)}$ and $t \in \mathcal{D}^{\theta(s,p)}$ such that $s+t \in \mathcal{D}^{(p)}$, we have $\theta(s, \theta(t,p)) = \theta(s+t,p)$*

Remark. *For each $p \in M$, define a curve $\theta^{(p)} : \mathcal{D}^{(p)} \to M$ by $\theta^{(p)}(t) = \theta(t,p)$*

Definition 3.9. *Given a smooth flow θ on M, the **infinitesimal generator** of θ is the vector field X defined by*

$$p \mapsto \theta^{(p)\prime}(0)$$

Proposition 3.10. *The infinitesimal generator X of θ is smooth, and each curve $\theta^{(p)}$ is an integral curve of X.*

Theorem 3.11 (Fundamental Theorem on Flows)**.**

Chapter III

Vector Bundles

§1 Vector Bundles

Definition 1.1. *Let M be a topological space. A* ***vector bundle of rank*** k ***over*** M *is a topological space E together with a continuous surjective map*

$$\pi : E \to M$$

such that

(i) for each $p \in M$, the fiber $E_p = \pi^{-1}(p)$ is a k-dimensional real vector space;

(ii) every $p \in M$ has an open neighborhood U and a diffeomorphism

$$\Phi : \pi^{-1}(U) \to U \times \mathbb{R}^k$$

satisfying $\pi_1 \circ \Phi = \pi$, where $\pi_1 : U \times \mathbb{R}^k \to U$ is projection;

(iii) for each $q \in U$, the restricted map

$$\Phi|_{E_q} : E_q \to \{q\} \times \mathbb{R}^k \cong \mathbb{R}^k$$

is a linear isomorphism.

The manifold E is the ***total space****, M is the* ***base****, π is the* ***projection****, and Φ is a* ***smooth local trivialization****.*

Example 1.2. *The tangent bundle $TM \to M$ is a rank-n vector bundle when* $\dim M = n$*. If $(U, x^1, \ldots, x^n)$ is a smooth chart, then*

$$\Phi\left(v^i \frac{\partial}{\partial x^i}\bigg|_p\right) = (p, (v^1, \ldots, v^n))$$

is a local trivialization of TM over U.

Definition 1.3. *If $\Phi : \pi^{-1}(U) \to U \times \mathbb{R}^k$ and $\Psi : \pi^{-1}(V) \to V \times \mathbb{R}^k$ are local trivializations of a rank-k vector bundle, then on $U \cap V$ their* ***transition function*** *is the smooth map*

$$\tau : U \cap V \to \mathrm{GL}(k, \mathbb{R})$$

defined by

$$\Psi \circ \Phi^{-1}(p, v) = (p, \tau(p)v).$$

Thus changes of vector bundle coordinates are linear on each fiber, and the corresponding linear maps vary smoothly with the base point.

Lemma 1.4 (Vector Bundle Chart Lemma). *Let M be a smooth manifold and let $\pi : E \to M$ be a surjective map whose fibers are k-dimensional vector spaces. Suppose there is an open cover $\{U_\alpha\}$ of M and bijections*

$$\Phi_\alpha : \pi^{-1}(U_\alpha) \to U_\alpha \times \mathbb{R}^k$$

such that $\pi_1 \circ \Phi_\alpha = \pi$, each $\Phi_\alpha|_{E_p}$ is a linear isomorphism, and the transition maps have the form

$$\Phi_\beta \circ \Phi_\alpha^{-1}(p, v) = (p, \tau_{\beta\alpha}(p)v)$$

with $\tau_{\beta\alpha} : U_\alpha \cap U_\beta \to \mathrm{GL}(k, \mathbb{R})$ smooth. Then E has a unique smooth manifold structure for which $\pi : E \to M$ is a smooth vector bundle and the Φ_α are smooth local trivializations.

Example 1.5 (Whitney Sum). *If $\pi' : E' \to M$ and $\pi'' : E'' \to M$ are smooth vector bundles of ranks k' and k'', their* ***Whitney sum*** *is*

$$E' \oplus E'' = \coprod_{p \in M} \left(E'_p \oplus E''_p\right).$$

Using local trivializations of E' and E'' over the same open set gives local trivializations

$$(E' \oplus E'')|_U \cong U \times \mathbb{R}^{k'+k''}.$$

Hence $E' \oplus E''$ is a smooth vector bundle of rank $k' + k''$.

Example 1.6 (Restriction). *If $\pi : E \to M$ is a smooth vector bundle and $S \subseteq M$ is an embedded submanifold, then*

$$E|_S = \pi^{-1}(S)$$

is a smooth vector bundle over S with projection $\pi|_{E|_S} : E|_S \to S$. Its fiber over $p \in S$ is still E_p.

§2 Local and Global Sections

Definition 2.1. *Let $\pi : E \to M$ be a vector bundle.*

1. *A **global section** of E is a continuous map $\sigma : M \to E$ such that $\pi \circ \sigma = \mathrm{id}_M$.*

   *A **local section** of E is a continuous map $\sigma : U \to E$ such that $\pi \circ \sigma = \mathrm{id}_U$.*

2.

Definition 2.2. *Let $E \to M$ be a vector bundle and a open subset $U \subseteq M$. A **local frame** for E over U is a k-tuple of local sections*

$$(\sigma_1, \ldots, \sigma_k)$$

*such that $(\sigma_1(p), \ldots, \sigma_k(p))$ is a basis for E_p for every $p \in U$. A frame over all of M is a **global frame**.*

Proposition 2.3 (Frames and Trivializations)**.** *Smooth local trivializations and smooth local frames determine each other. If $\Phi : \pi^{-1}(U) \to U \times \mathbb{R}^k$ is a local trivialization and $(e_1, \ldots, e_k)$ is the standard basis of $\mathbb{R}^k$, then*

$$\sigma_i(p) = \Phi^{-1}(p, e_i)$$

is a smooth local frame over U. Conversely, if $(\sigma_1, \ldots, \sigma_k)$ is a smooth local frame, then

$$\Phi\left(a^i \sigma_i(p)\right) = (p, (a^1, \ldots, a^k))$$

defines a smooth local trivialization.

Corollary 2.4. *A smooth rank-k vector bundle is smoothly trivial if and only if it admits a smooth global frame.*

Proposition 2.5 (Completion of Local Frames)**.** *Let $E \to M$ be a smooth vector bundle of rank k. If smooth local sections $\sigma_1, \ldots, \sigma_j$ are linearly independent at a point p, then after shrinking to a neighborhood U of p they can be completed to a smooth local frame*

$$(\sigma_1, \ldots, \sigma_j, \sigma_{j+1}, \ldots, \sigma_k)$$

for E over U.

Corollary 2.6. *Given $p \in M$ and $v \in E_p$, there is a smooth local section σ of E defined near p such that $\sigma(p) = v$. Using a bump function, one may choose σ to extend to a smooth global section whose support lies in a prescribed coordinate neighborhood.*

Proposition 2.7 (Local Frame Criterion for Smoothness)**.** *Let $(\sigma_1, \ldots, \sigma_k)$ be a smooth local frame for E over U. A local section $\sigma : U \to E$ is smooth if and only if the component functions $a^1, \ldots, a^k$ in*

$$\sigma = a^i \sigma_i$$

are smooth functions on U.

§3 Bundle Homomorphisms

Definition 3.1. *Let $\pi : E \to M$ and $\pi' : E' \to M'$ be smooth vector bundles. A* ***bundle homomorphism*** *from E to E' is a pair of smooth maps*

$$F : E \to E', \qquad f : M \to M'$$

such that

$$\pi' \circ F = f \circ \pi$$

and each restricted map

$$F|_{E_p} : E_p \to E'_{f(p)}$$

is linear. If $M = M'$ and $f = \mathrm{id}_M$, then F is said to be a bundle homomorphism ***over*** *M.*

Definition 3.2. *A bundle homomorphism $F : E \to E'$ over M is a* ***bundle isomorphism*** *if it is bijective and its inverse is also a smooth bundle homomorphism over M. Two vector bundles over M are* ***smoothly isomorphic*** *if there exists such an isomorphism between them.*

Proposition 3.3. *A bijective smooth bundle homomorphism over M is a smooth bundle isomorphism.*

Proposition 3.4 (Matrix Criterion)**.** *Let $E, E' \to M$ be smooth vector bundles and let $F : E \to E'$ be a bundle homomorphism over M. If $(\sigma_1, \dots, \sigma_k)$ is a smooth local frame for E and $(\tau_1, \dots, \tau_\ell)$ is a smooth local frame for E' over U, then*

$$F(\sigma_j) = F^i{}_j \, \tau_i.$$

The map F is smooth if and only if all coefficient functions $F^i{}_j$ are smooth on U for every such choice of local frames.

Lemma 3.5 (Characterization by Sections)**.** *Let $E, E' \to M$ be smooth vector bundles. A smooth bundle homomorphism $F : E \to E'$ over M induces a $C^\infty(M)$-linear map*

$$\Gamma(E) \to \Gamma(E'), \qquad \sigma \mapsto F \circ \sigma.$$

Conversely, every $C^\infty(M)$-linear map $\Gamma(E) \to \Gamma(E')$ is induced by a unique smooth bundle homomorphism over M.

Example 3.6. *If $f \in C^\infty(M)$, scalar multiplication*

$$v \mapsto f(\pi(v))v$$

is a smooth bundle homomorphism $E \to E$ over M. On sections it is the map $\sigma \mapsto f\sigma$.

Example 3.7. *If $F : M \to N$ is smooth, then the differential*

$$dF : TM \to TN$$

is a smooth bundle homomorphism covering F. On each fiber it is the linear map

$$dF_p : T_pM \to T_{F(p)}N.$$

§4 Subbundles

Definition 4.1. *Let $\pi : E \to M$ be a smooth vector bundle. A subset $D \subseteq E$ is a* ***smooth subbundle of rank*** *r if D is an embedded submanifold of E, the restriction $\pi|_D : D \to M$ is a smooth vector bundle of rank r, and each fiber*

$$D_p = D \cap E_p$$

is a vector subspace of E_p.

Lemma 4.2 (Local Frame Criterion for Subbundles). *Let $D \subseteq E$ be a subset such that each $D_p = D \cap E_p$ is an r-dimensional vector subspace of E_p. Then D is a smooth subbundle of E if and only if every $p \in M$ has a neighborhood U on which there exist smooth local sections $\sigma_1, \dots, \sigma_r$ of E satisfying*

$$D_q = \operatorname{span}(\sigma_1(q), \dots, \sigma_r(q))$$

for all $q \in U$.

Example 4.3. *If $S \subseteq M$ is an embedded submanifold, then TS is naturally a smooth subbundle of $TM|_S$. At each point $p \in S$,*

$$T_pS \subseteq T_pM.$$

Theorem 4.4 (Kernel and Image Subbundles). *Let $F : E \to E'$ be a smooth bundle homomorphism over M. If the rank of the linear map*

$$F_p : E_p \to E'_p$$

is constant on M, then

$$\operatorname{Ker} F = \coprod_{p \in M} \operatorname{Ker} F_p \qquad \textit{and} \qquad \operatorname{Im} F = \coprod_{p \in M} \operatorname{Im} F_p$$

are smooth subbundles of E and E' respectively.

Definition 4.5. *If $M \subseteq \mathbb{R}^n$ is an embedded submanifold, its* ***normal bundle*** *is*

$$NM = \coprod_{p \in M} N_pM, \qquad N_pM = (T_pM)^\perp \subseteq T_p\mathbb{R}^n \cong \mathbb{R}^n.$$

Corollary 4.6. *If $M \subseteq \mathbb{R}^n$ is an embedded submanifold, then NM is a smooth subbundle of $T\mathbb{R}^n|_M$. Moreover*

$$T\mathbb{R}^n|_M = TM \oplus NM.$$

Remark. *The central principle of vector bundles is local triviality: near each point, a vector bundle looks like $U \times \mathbb{R}^k$, but the transition functions can glue these local products nontrivially. Sections generalize vector fields, frames are the bundle analogue of bases, and bundle homomorphisms are fiberwise linear maps varying smoothly with the base point.*

Chapter IV

Tensors

§1

Given a vector space V, denote by $T^k(V)$ the k-th tensor power $V^{\otimes k}$. And view the $T^k(V^*) \cong T^k(V)^* := \operatorname{Hom}_{\mathbb{R}}\left(V^{\otimes k}, \mathbb{R}\right)$ as the space of multilinear maps from V^k to $\mathbb{R}$ (with the convention that $T^0(V^*) = \mathbb{R}$).

Definition 1.1. *1. The* ***tensor algebra*** *of V^* is the direct sum of all tensor powers of V^*:*

$$T(V^*) := \bigoplus_{k=0}^{\infty} T^k(V^*)$$

with the

Definition 1.2. *Given a k-covector $\omega \in T^k(V^*)$, we define*

1. *the* ***symmetrization*** *of ω to be the tensor*

$$\operatorname{Sym}(\omega)(v_1, \ldots, v_k) := \frac{1}{k!} \sum_{\sigma \in S_k} \omega\left(v_{\sigma(1)}, \ldots, v_{\sigma(k)}\right)$$

2. *the* ***antisymmetrization*** *of ω to be the tensor*

$$\operatorname{Alt}(\omega)(v_1, \ldots, v_k) := \frac{1}{k!} \sum_{\sigma \in S_k} \operatorname{sgn}(\sigma)\omega\left(v_{\sigma(1)}, \ldots, v_{\sigma(k)}\right)$$

Definition 1.3. *Given $\omega \in T^k(V^*)$ and $\eta \in T^l(V^*)$. The* ***wedge product*** *of ω and η is defined to be*

$$\omega \wedge \eta := \frac{(k+l)!}{k!l!} \operatorname{Alt}(\omega \otimes \eta)$$

The ***symmetric product*** *of ω and η is defined to be*

$$\omega \odot \eta := \frac{(k+l)!}{k!l!} \operatorname{Sym}(\omega \otimes \eta)$$

Proposition 1.4. *The wedge product has the following properties:*

1. *associative*
$$(\omega \wedge \eta) \wedge \xi = \omega \wedge (\eta \wedge \xi) = \frac{(k+l+m)!}{k!l!m!} \operatorname{Alt}(\omega \otimes \eta \otimes \xi)$$
2. *bilinear*
3. *Anti-commutative relation:* $\omega \wedge \eta = (-1)^{kl} \eta \wedge \omega$
4. *For any* $f^i \in V^*$ *and* $v^i \in V$*, we have*
$$f^1 \wedge \cdots \wedge f^k (v_1, \ldots, v_k) = \det\left(f^i(v_i)\right)$$

Corollary 1.5. *Given a basis* $\{f^1, \ldots, f^n\}$ *for* V^**, the set* $\{f^{i_1} \wedge \cdots \wedge f^{i_k} | 1 \leq i_1 < \cdots < i_k \leq n\}$ *is a basis for* $\Lambda^k(V^*)$ *.*

Definition 1.6. 1. *The* ***exterior algebra*** *of* V^* *is the graded algebra*
$$\Lambda(V^*) := \bigoplus_{k=0}^{\infty} \Lambda^k(V^*)$$
with the wedge product as multiplication.

2. *The* ***symmetric algebra*** *is the graded algebra*
$$\operatorname{Sym}(V^*) := \bigoplus_{k=0}^{\infty} \operatorname{Sym}^k(V^*)$$
with the symmetric product as multiplication.

Definition 1.7. *Let* V *be a vector space. A* (k,l)*-tensor* T *on* V *is a element of the vector space*
$$T^{k,l}(V) := T^k(V) \otimes T^l(V^*)$$

Remark. *A multilinear map* $T : \underbrace{V^* \times \cdots \times V^*}_{k} \times \underbrace{V \times \cdots \times V}_{l} \to \mathbb{R}$.

§2 Tensor Fields

Definition 2.1. *The tnesor space of type* (k,l) *at* p *is the vector space*
$$T_p^{(k,l)} M := T^k(T_pM) \otimes T^l\left(T_p^* M\right)$$

The ***tensor bundle*** *of type* (k,l) *over* M *is the vector bundle*
$$T^{(k,l)} M := \coprod_{p \in M} T_p^{(k,l)} M =$$

*The **tensor fields** of type (k,l) on M are the smooth sections of $T^{(k,l)}M \to M$.*

Remark. *Given a smooth chart (U, x^i), we can write the (k,l)-tensor fields A (omit the base point) in coordinates locally as*

$$A = A^{i_1 \dots i_k}_{j_1 \dots j_l} \cdot \frac{\partial}{\partial x^{i_1}} \otimes \cdots \otimes \frac{\partial}{\partial x^{i_k}} \otimes dx^{j_1} \otimes \cdots \otimes dx^{j_l}$$

Pullback and Lie Derivative of Tensor Fields

Definition 2.2. *Suppose $f : M \to N$ is a smooth map and B is k-covariant tensor field on N. The **pullback** of B by f is the k-covariant tensor field f^*B on M defined pointwise by*

$$(f^*B)_p \left(X_1, \dots, X_k\right) := B_{f(p)} \left(f_{*,p}X_1, \dots, f_{*,p}X_k\right)$$

for all $p \in M$ and $X_1, \dots, X_k \in T_pM$.

Proposition 2.3.

$$f^*(B_{i_1 \dots i_k} dy^{i_1} \otimes \cdots \otimes dy^{i_k}) = B_{i_1 \dots i_k} \circ f \cdot d(y^{i_1} \circ f) \otimes \cdots \otimes d(y^{i_k} \circ f)$$

Proposition 2.4. *The pullback is a contravariant functor from the category of smooth manifolds to the category of graded algebras:*

$$\begin{array}{ccc} M & \mapsto & T^{(0,*)}M \\ {\scriptstyle f}\big\downarrow & \mapsto & \big\uparrow{\scriptstyle f^*} \\ N & \mapsto & T^{(0,*)}N \end{array}$$

where f^* is a graded algebra homomorphism, i.e.

1. *$f^*(A \otimes B) = f^*A \otimes f^*B$*
2. *$f^*(A + B) = f^*A + f^*B$ and $f^*(\lambda A) = \lambda f^*A$ for all $\lambda \in \mathbb{R}$.*

Corollary 2.5. 1. *$F^*(fB) = f \circ F \cdot F^*B$*

2. *The pullbak also commutes with the symmetrization, antisymmetrization, wedge product and symmetric product operations.*

Definition 2.6. *The Lie derivative of a tensor field A with respect to a vector field X is defined by*

$$\mathcal{L}_X A := \lim_{t \to 0} \frac{(\theta_t)^*_p A_{\theta_t(p)} - A_p}{t}$$

§3 Differential Forms

Proposition 3.1. *If $\omega \in \Lambda^n(V^*)$ and $T : V \to V$ is linear, then*

$$\omega(Tv_1, \ldots, Tv_n) = (\det T)\omega(v_1, \ldots, v_n).$$

Thus top-degree alternating tensors transform by determinants.

Definition 3.2. *For $X \in V$, the **interior multiplication** or **contraction** by X is the map*

$$\iota_X : \Lambda^k(V^*) \to \Lambda^{k-1}(V^*)$$

defined by

$$(\iota_X\omega)(X_1, \ldots, X_{k-1}) = \omega(X, X_1, \ldots, X_{k-1}).$$

Proposition 3.3. *Interior multiplication is an antiderivation of degree -1 on the exterior algebra:*

$$\iota_X(\omega \wedge \eta) = (\iota_X\omega) \wedge \eta + (-1)^k\omega \wedge (\iota_X\eta), \qquad \omega \in \Lambda^k(V^*).$$

Definition 3.4. *Let M be a smooth manifold, possibly with boundary.*

1. *The bundle of alternating k-covectors is*

$$\Lambda^kT^*M = \coprod_{p\in M} \Lambda^k(T_p^*M).$$

2. *A **differential** k**-form** on M is a smooth section of $\Lambda^k(T^*M) \to M$. The space of all smooth k-forms is denoted by*

$$\Omega^k(M) = \Gamma(\Lambda^kT^*M).$$

   By convention, $\Omega^0(M) = \Gamma(M \times \mathbb{R}) = C^\infty(M)$.

3. *If $\omega \in \Omega^k(M)$ and $\eta \in \Omega^l(M)$, their wedge product is defined pointwise:*

$$(\omega \wedge \eta)_p = \omega_p \wedge \eta_p.$$

   Thus $\omega \wedge \eta \in \Omega^{k+l}(M)$.

Remark. *In a smooth chart $(U, x^1, \ldots, x^n)$, every k-form ω can be written uniquely as*

$$\omega = \sum_{i_1<\cdots<i_k} \omega_{i_1\cdots i_k}dx^{i_1} \wedge \cdots \wedge dx^{i_k} = \sum_I \omega_I\, dx^{i_1} \wedge \cdots \wedge dx^{i_k},$$

where the coefficient functions $\omega_{i_1\cdots i_k}$ are smooth on U and multiindex $I = (i_1, \ldots, i_k)$.

§4 Inner Products, Lie Derivatives and Exterior Derivatives

Definition 4.1. *If X is a smooth vector field on M and $\omega \in \Omega^k(M)$, the interior product*

$$\iota_X \omega \in \Omega^{k-1}(M)$$

is defined by

$$(\iota_X \omega)_p = \iota_{X_p}(\omega_p).$$

Definition 4.2. *The **exterior derivative** is the unique map*

$$d : \Omega^*(M) \to \Omega^*(M)$$

characterized by the following properties:

(1) if $f \in C^\infty(M)$, then df is the ordinary differential of f;

(2) d is $\mathbb{R}$-linear;

(3) $d(\omega \wedge \eta) = d\omega \wedge \eta + (-1)^k \omega \wedge d\eta$ for $\omega \in \Omega^k(M)$;

(4) $d(d\omega) = 0$ for every differential form ω.

Proposition 4.3. *In local coordinates $(x^1, \dots, x^n)$, if*

$$\omega = \sum_I \omega_I \, dx^{i_1} \wedge \cdots \wedge dx^{i_k},$$

then

$$d\omega = \sum_I d\omega_I \wedge dx^{i_1} \wedge \cdots \wedge dx^{i_k} = \sum_{j,I} \frac{\partial \omega_I}{\partial x^j} dx^j \wedge dx^{i_1} \wedge \cdots \wedge dx^{i_k}.$$

Definition 4.4. *A form ω is called **closed** if $d\omega = 0$. It is called **exact** if there exists a form η such that $\omega = d\eta$*

Remark. *Since $d^2 = 0$, every exact form is closed.*

Proposition 4.5 (Invariant Formula for d). *Let $\omega \in \Omega^k(M)$ and let $X_0, \dots, X_k$ be smooth vector fields. Then*

$$\begin{aligned} d\omega(X_0, \dots, X_k) &= \sum_{i=0}^{k} (-1)^i X_i\big(\omega(X_0, \dots, \widehat{X_i}, \dots, X_k)\big) \\ &\quad + \sum_{0 \le i < j \le k} (-1)^{i+j} \omega([X_i, X_j], X_0, \dots, \widehat{X_i}, \dots, \widehat{X_j}, \dots, X_k). \end{aligned}$$

Here a hat means that the indicated argument is omitted.

Theorem 4.6 (Cartan's Magic Formula)**.** *On a smooth manifold, for every smooth vector field X and every differential form ω,*

$$\mathcal{L}_X\omega = \iota_X(d\omega) + d(\iota_X\omega).$$

Proposition 4.7. 1. *The Lie derivative commutes with the exterior derivative:*

$$\mathcal{L}_X(d\omega) = d(\mathcal{L}_X\omega).$$

2. *Exterior differentiation commutes with pullback*

$$f^*(d\omega) = d(f^*\omega)$$

3. *The Lie derivative commutes with pullback:*

Chapter V

Submersions, Immersions, and Embeddings

In this chapter, manifolds are assumed to be smooth manifolds with or without boundary unless otherwise stated.

§1

Theorem 1.1 (Rank Theorem)**.** *Suppose M and N are smooth manifolds of dimensions m and n, respectively, and $f : M \to N$ is a smooth map with constant rank r. For each $p \in M$ there exist smooth charts (U, φ) for M centered at p and (V, ψ) for N centered at $f(p)$ such that $f(U) \subseteq V$, in which f has a coordinate representation of the form*

$$\hat{f}(x^1, \ldots, x^r, x^{r+1}, \ldots, x^m) = (x^1, \ldots, x^r, 0, \ldots, 0).$$

§2 Submersions

Theorem 2.1 (Local Section Theorem)**.** *The smooth map $\pi : M \to N$ is a submersion if and only if every point $p \in M$ is in the image of a smooth local section of π.*

§3 Embeddings

Lemma 3.1 (Transport of structure)**.** *Let $k \in \mathbb{N}$ and $f : M \to N$ be a homeomorphism between C^k-manifold M and a topological space N. Then there is a unique C^k-structure on N such that f is a C^k-diffeomorphism.*

Remark. *If $f : M \to N$ is a homeomorphism between manifolds, f is not necessarily a diffeomorphism. For example, the map $f : \mathbb{R} \to \mathbb{R}$ defined by $f(x) = x^3$ is a homeomorphism but not a diffeomorphism.*

Definition 3.2. *A smooth map $f : M \to N$ is a* ***smooth embedding*** *if it is an immersion and a topological embedding. In this case, we say that M is* ***smoothly embedded*** *in N, and $S = f(M) \cong M$ is an* ***embedded submanifold*** *of N.*

Remark. *By the transport of structure lemma, there is a unique smooth (topological) structure on S such that $f : M \to S$ is a diffeomorphism (homeomorphism). And the structure on S is compatible with N, meaning that the inclusion $S \hookrightarrow N$ is a smooth (topological) embedding.*

The Whitney Embedding Theorem

Theorem 3.3 (Whitney Embedding Theorem)**.** *Every smooth n-manifold with or without boundary admits a proper smooth embedding into $\mathbb{R}^{2n+1}$.*

Corollary 3.4. *Every smooth n-dimensional manifold with or without boundary is diffeomorphic to a properly embedded submanifold (with or without boundary) of $\mathbb{R}^{2n+1}$.*

Corollary 3.5. *Suppose M is a compact smooth n-manifold with or without boundary. If $N \geq 2n+1$, then every smooth map from M to $\mathbb{R}^N$ can be uniformly approximated by embeddings.*

Theorem 3.6 (Whitney Immersion Theorem)**.** *Every smooth n-manifold with or without boundary admits a smooth immersion into $\mathbb{R}^{2n}$.*

Theorem 3.7 (Strong Whitney Immersion Theorem)**.** *If $n > 1$, every smooth n-manifold admits a smooth immersion into $\mathbb{R}^{2n-1}$.*

§4 Submanifold

Definition 4.1. *An **embedded submanifold** of M is a subset $S \subseteq M$ such that*

1. *S is a manifold*
2. *the inclusion map $i : S \hookrightarrow M$ is a smooth embedding.*

Definition 4.2. *Given an subset $S \subset M$, $k \in \mathbb{N}$ and embedding $i : S \hookrightarrow M$.*

1. *If there is a chart (U, φ) covering S and $\varphi(S) = \{x \in \varphi(U) | x^i = 0, i = k+1, \ldots, m\}$, then we say that S is a k-**slice** of U.*
2. *we say that S satisfies the **local** k**-slice condition** if each point of S is contained a chart (U, φ) for M such that $S \cap U$ is a k-slice in U.*

*Any such chart is called a **slice chart** for S in M, and the corresponding coordinates $(x^1, \ldots, x^n)$ are called **slice coordinates**.*

Theorem 4.3 (Local Slice Criterion for Embedded Submanifolds)**.** *If $S \subset M$ is an embedded k-dimensional submanifold, then S satisfies the local k-slice condition.*

Theorem 4.4. *If M is a smooth n-manifold with boundary, then ∂M is an embedded $(n-1)$-dimensional submanifold of M.*

Theorem 4.5. *Let $f : M \to N$ be a smooth map with constant rank r. Then each level set $f^{-1}(c)$ is a properly embedded submanifold of codimension r in M.*

Corollary 4.6. *Every regular level set of a smooth map is a properly embedded submanifold whose codimension is equal to the dimension of the codomain.*